Boundary-Corrected Longitudinal Modular Reduction for a Wilson–Dirac Interval

Kirtus Justus (Theoretical Physicist) • 31 October 2026
Port Coquitlam, British Columbia, Canada

Abstract

For a finite interval cut from the infinite Wilson–Dirac occupied-band projection, the longitudinal perturbation can be rewritten exactly as a positive-majorant-weighted reference term plus a cross-boundary correction. The correction admits an explicit Hilbert–Schmidt bound controlled by the occupation-weighted modular quadratic form. As a consequence, the leading-order relative-entropy reduction can be equivalently formulated using the positive majorant and the reference occupation operator. This argument reformulates the modular estimate. The separate whole-line argument in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* controls that estimate and establishes the leading entropy law.

1. Operator setting

Let $\mathcal{H} = \ell^2(\mathbb{Z}) \otimes \mathbb{C}^2$ and let χ be the orthogonal projection onto L consecutive two-orbital cells, $N = 2L$. Operators on $\operatorname{ran}\chi$ are implicitly restricted to that space. All full-line operators are expressed after one common, interval-preserving cellwise rotation and staggering.

Let P_0 be the massless occupied-band projection and G the positive Wilson spectral majorant. For fixed $\mu > 0$ and $m = \mu/L$, the full-line projector calculation gives

$$P_0^2 = P_0 = P_0^*, \qquad G = G^*, \qquad 0 \le G \le \tfrac{1}{2} I, \qquad [G, P_0] = 0,$$
$$S = G(I - 2P_0), \qquad \operatorname{Tr}(\chi G \chi) \le \frac{\mu}{\pi}. \tag{1}$$

Define

$$B_N = \chi P_0 \chi, \qquad G_N = \chi G \chi, \qquad S_N = \chi S \chi, \qquad W_N = B_N(I - B_N). \tag{2}$$

The compressed reference satisfies $0 < B_N < I$. Its commuting Jacobi operator J_N has off-diagonal entries $j(N-j)/N$. Put

$$K_N = \log(I - B_N) - \log B_N, \quad E_N = K_N + N\eta(J_N/N), \tag{3}$$

with the half-filled reference scaling function η from the free-fermion Jacobi model. In particular, $E_N = E_N^*$ and $[E_N, B_N] = 0$. The occupation-weighted estimate is

$$\operatorname{Tr}(W_N E_N^2) \le C(1 + \log N) \quad (N \ge 2), \tag{4}$$

with a constant independent of N. We explicitly retain (1) and (4) as input hypotheses to the new theorem.

2. Exact boundary-corrected decomposition

Theorem 1. Let $Q = I - \chi$ and set

$$C_N = \chi G Q P_0 \chi.$$

Then for every finite interval,

$$\boxed{S_N = G_N - G_N B_N - B_N G_N - C_N - C_N^*.} \tag{5}$$

Consequently,

$$\boxed{\operatorname{Tr}(S_N E_N) = \operatorname{Tr}\big[G_N (I - 2B_N) E_N\big] - 2 \operatorname{Re} \operatorname{Tr}(C_N E_N).} \tag{6}$$

Proof. From $S = G(I - 2P_0)$ and $[G, P_0] = 0$,

$$S_N = G_N - \chi(G P_0 + P_0 G)\chi.$$

Insert $I = \chi + Q$ between the two full-line operators in each product:

$$\begin{aligned} \chi G P_0 \chi &= G_N B_N + \chi G Q P_0 \chi, \\ \chi P_0 G \chi &= B_N G_N + \chi P_0 Q G \chi. \end{aligned}$$

Because G, P_0 are self-adjoint, the last term is C_N^*. This proves (5), without assuming that G_N commutes with B_N. Multiply by E_N, trace, use $[E_N, B_N] = 0$ and cyclicity, and combine the conjugate boundary traces to obtain (6). □

3. Sublogarithmic boundary control

Theorem 2. Under the hypotheses (1)–(4),

$$\boxed{2\,|\operatorname{Tr}(C_N E_N)| \le \sqrt{\frac{2\mu}{\pi}}\,\big[\operatorname{Tr}(W_N E_N^2)\big]^{1/2} = O_\mu(\sqrt{\log N}).} \tag{7}$$

Proof. The two factors in $C_N = (QG\chi)^*(QP_0\chi)$ are Hilbert–Schmidt after restriction to the finite interval. Because $0 \le Q \le I$ and $G^2 \le G/2$,

$$\begin{aligned} \|QG\chi\|_{\mathrm{HS}}^2 &= \operatorname{Tr}(\chi G Q G \chi) \\ &\le \operatorname{Tr}(\chi G^2 \chi) \le \tfrac{1}{2} \operatorname{Tr} G_N \le \frac{\mu}{2\pi}. \end{aligned} \tag{8}$$

The full-line projection identity gives

$$\chi P_0 Q P_0 \chi = \chi P_0 \chi - (\chi P_0 \chi)^2 = W_N,$$

and hence

$$\|Q P_0 \chi E_N\|_{\mathrm{HS}}^2 = \operatorname{Tr}(E_N W_N E_N) = \operatorname{Tr}(W_N E_N^2). \tag{9}$$

Cauchy–Schwarz for the Hilbert–Schmidt inner product, (8), (9), and (4) prove the assertion. □

The exact Hilbert–Schmidt factor is the boundary leakage $\Lambda_N = \operatorname{Tr}(\chi G Q G \chi) = \operatorname{Tr}(\chi G^2 \chi) - \operatorname{Tr}(G_N^2)$. Consequently, the proof also gives

$$2|\operatorname{Tr}(C_N E_N)| \le 2\sqrt{\Lambda_N}\,[\operatorname{Tr}(W_N E_N^2)]^{1/2}.$$

For fixed $\mu > 0$, the companion theorem *Continuum Limit of the Wilson Majorant Boundary Defect* establishes $\Lambda_N \to \Phi(\mu) > 0$. This identifies the asymptotic Hilbert–Schmidt prefactor without changing the $O_\mu(\sqrt{\log N})$ growth order.

4. Equivalent leading-order obstruction

The established four-branch entropy decomposition and its auxiliary estimates give, for fixed $\mu > 0$,

$$\mathcal{D}_L(\mu) = \frac{\mu^2}{6}\log L + \operatorname{Tr}(S_N E_N) + O_\mu(\sqrt{\log L}), \qquad N = 2L. \tag{10}$$

Corollary 3. Under the same hypotheses,

$$\boxed{\mathcal{D}_L(\mu) = \frac{\mu^2}{6}\log L + \operatorname{Tr}\big[G_N(I - 2B_N)E_N\big] + O_\mu(\sqrt{\log L}).} \tag{11}$$

In a common orthonormal eigenbasis $B_N v_j = b_j v_j$, $J_N v_j = \lambda_j v_j$, with $E_N v_j = e_j v_j$ and $q_j = \langle v_j, G_N v_j\rangle \ge 0$, the remaining bulk trace is exactly

$$\boxed{\operatorname{Tr}\big[G_N(I - 2B_N)E_N\big] = \sum_{j=1}^{N} q_j(1 - 2b_j)e_j.} \tag{12}$$

Thus the original condition $\operatorname{Tr}(S_N E_N) = o(\log N)$ is **equivalent** to the signed bulk condition

$$\sum_j q_j(1 - 2b_j)e_j = o(\log N). \tag{13}$$

A sufficient noncancellation criterion is

$$\boxed{\sum_j q_j|1 - 2b_j|\,|e_j| = o(\log N).} \tag{14}$$

It is no stronger than the stronger sufficient bound $\sum_j q_j|e_j| = o(\log N)$, since $|1 - 2b_j| \le 1$; the extra occupation factor suppresses contributions from reference levels near half-filling. The separate weighted theorem in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* proves the stronger bound $\sum_j q_j|e_j| = O_\mu(1)$, and hence both conditions at fixed positive scaled mass. Numerical tests alone would not prove either condition.

5. Scope and limitations

Equations (5) and (6) are exact at finite interval length. Equation (7) is a proved consequence of the explicitly identified Wilson-majorant and weighted-modular inputs. Equation (11) is a reduction with a rigorously bounded error; identifying its leading coefficient additionally uses the weighted theorem cited next. The bulk trace in (13) is controlled by the separate weighted theorem in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*. A finite entropy part and an inverse-logarithmic convergence rate do not follow from this reduction alone. For the specified interval restrictions with fixed positive scaled mass, Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy* supplies that limit and rate.

The interval is a compression of the *infinite-chain occupied-band projection*. None of the statements concerns ground-state projectors of a finite open Wilson chain unless additional arguments establish that identification. Operators on the full line are never replaced by products of their compressions without the boundary terms displayed above.

References and mathematical inputs

- D. Slepian, "Prolate spheroidal wave functions, Fourier analysis, and uncertainty V: The discrete case," *Bell System Technical Journal* **57** (1978), 1371–1430, DOI: 10.1002/j.1538-7305.1978.tb02104.x.
- V. Eisler and I. Peschel, "Analytical results for the entanglement Hamiltonian of a free-fermion chain," *Journal of Physics A* **50** (2017), 284003, DOI: 10.1088/1751-8121/aa76b5.
- K. Justus, "Exact Spectral Majorants for a Wilson–Dirac Interval," research manuscript (2026), for the model-specific full-line decomposition and trace majorant.
- K. Justus, "Logarithmic Weighted Modular Control for the Half-Filled Sine Kernel," research manuscript (2026), for the input (4).
- K. Justus, "Finite-Interval Wilson–Dirac Relative Entropy," research manuscript (2026), for the four-branch reduction (10).
- K. Justus, "Continuum Limit of the Wilson Majorant Boundary Defect" (2026), for the fixed-scaled-mass limit of the Hilbert–Schmidt factor.

Contributions and acknowledgments

Kirtus Justus directed the research program and supplied its mathematical starting point. ChatGPT (OpenAI) developed the displayed boundary decomposition, its Hilbert–Schmidt estimate, and the accompanying computational cross-checks; these derivations have not undergone independent external peer review. Claude (Anthropic), Gemini (Google), Grok (SpaceXAI), and DeepSeek (High-Flyer) are acknowledged as participants in the broader research effort without assigning them unverified contributions to this argument. No institutional affiliation, endorsement, or peer-review status is implied.